



Faculté des Sciences et Technologies (FST)

Rapport du travail de Laboratoire N° 8_Réseaux I

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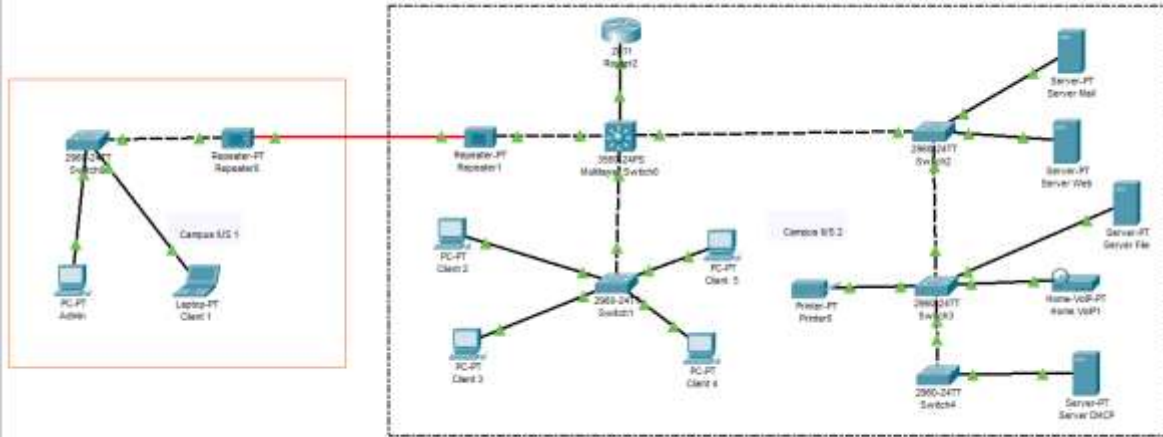
Professeur : Ismaël SAINT AMOUR

Niveau : L3

4 Janvier 2025

L'objectif de ce TD est de :

- Créer et gérer des VLAN sur des switches
- Comprendre le rôle des VLAN pour segmenter un réseau.
- Apprendre à créer des VLAN et à affecter des ports aux VLAN correspondants.
- Configurer le routage inter-VLAN sur un routeur (Router-on-a-Stick)
- Comprendre comment un routeur peut permettre aux VLAN de communiquer entre eux.
- Configurer des sous-interfaces avec encapsulation 802.1Q.
- Configurer les trunks entre switches et vers le routeur
- Apprendre à utiliser les trunks pour transporter plusieurs VLAN sur une seule liaison physique.
- Comprendre le concept de VLAN natif et VLAN autorisés sur les trunks.
- Attribuer des adresses IP et des gateways aux hôtes
- Configurer les PCs dans différents VLAN avec des IP correctes.
- Vérifier la communication intra-VLAN (même VLAN) et inter-VLAN (différents VLAN).



Vérifications importantes

- Vérification des vlans sur :



The screenshot shows a network switch configuration window titled "Switch1". The "CLI" tab is selected, displaying the "IOS Command Line Interface". The command "SW-R2>show vlan brief" has been executed, resulting in two identical tables showing the status and associated ports for various VLANs.

VLAN	Name	Status	Ports
1	default	active	Gig0/1, Gig0/2
10	Ventes	active	Fa0/2, Fa0/3, Fa0/4, Fa0/5 Fa0/6, Fa0/7
20	Marketing	active	Fa0/8, Fa0/9, Fa0/10, Fa0/11 Fa0/12, Fa0/13, Fa0/14, Fa0/15
30	IT	active	Fa0/16, Fa0/17, Fa0/18, Fa0/19 Fa0/20, Fa0/21, Fa0/22, Fa0/23 Fa0/24
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

Below the first table, the command "SW-R2>show vlan" is shown, followed by a second identical table.

VLAN	Name	Status	Ports
1	default	active	Gig0/1, Gig0/2
10	Ventes	active	Fa0/2, Fa0/3, Fa0/4, Fa0/5 Fa0/6, Fa0/7
20	Marketing	active	Fa0/8, Fa0/9, Fa0/10, Fa0/11 Fa0/12, Fa0/13, Fa0/14, Fa0/15
30	IT	active	Fa0/16, Fa0/17, Fa0/18, Fa0/19 Fa0/20, Fa0/21, Fa0/22, Fa0/23 Fa0/24
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

At the bottom of the CLI window, a header for a detailed VLAN table is visible:

VLAN	Type	SAID	MTU	Parent	RingNo	BridgeNo	Stp	BrdgMode	Trans1	Trans2
------	------	------	-----	--------	--------	----------	-----	----------	--------	--------

Switch2

PhysicalConfigCLIAttributes

IOS Command Line Interface

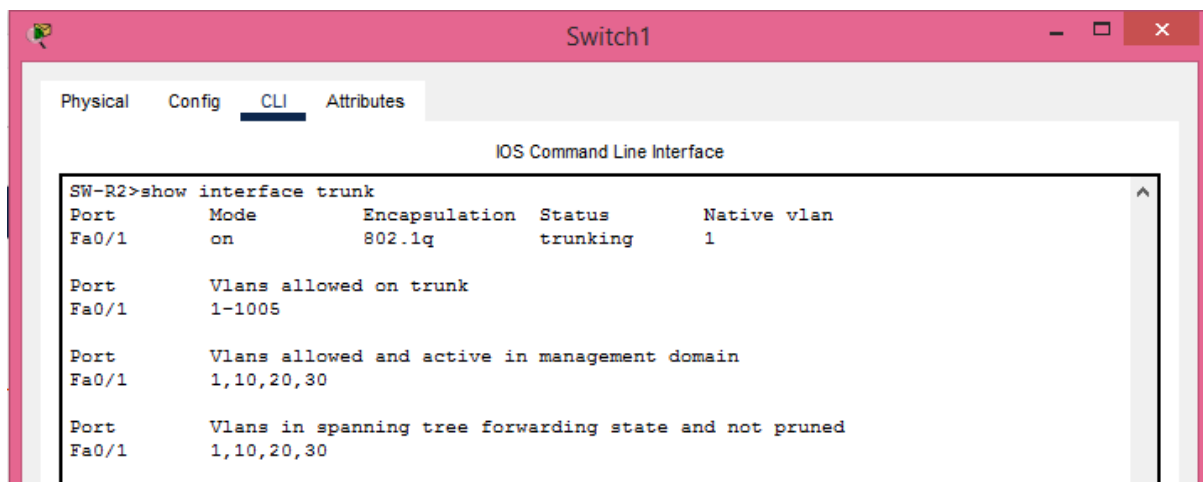
VLAN	Name	Status	Ports
1	default	active	Gig0/1, Gig0/2
10	Ventes	active	Fa0/2, Fa0/3, Fa0/4, Fa0/5 Fa0/6, Fa0/7
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1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

SW-R2>show vlan

VLAN	Name	Status	Ports
1	default	active	Gig0/1, Gig0/2
10	Ventes	active	Fa0/2, Fa0/3, Fa0/4, Fa0/5 Fa0/6, Fa0/7
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1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

VLAN	Type	SAID	MTU	Parent	RingNo	BridgeNo	Stp	BrdgMode	Trans1	Trans2
1	enet	100001	1500	-	-	-	-	-	0	0

- Vérification des trunks :



Switch1

Physical Config CLI Attributes

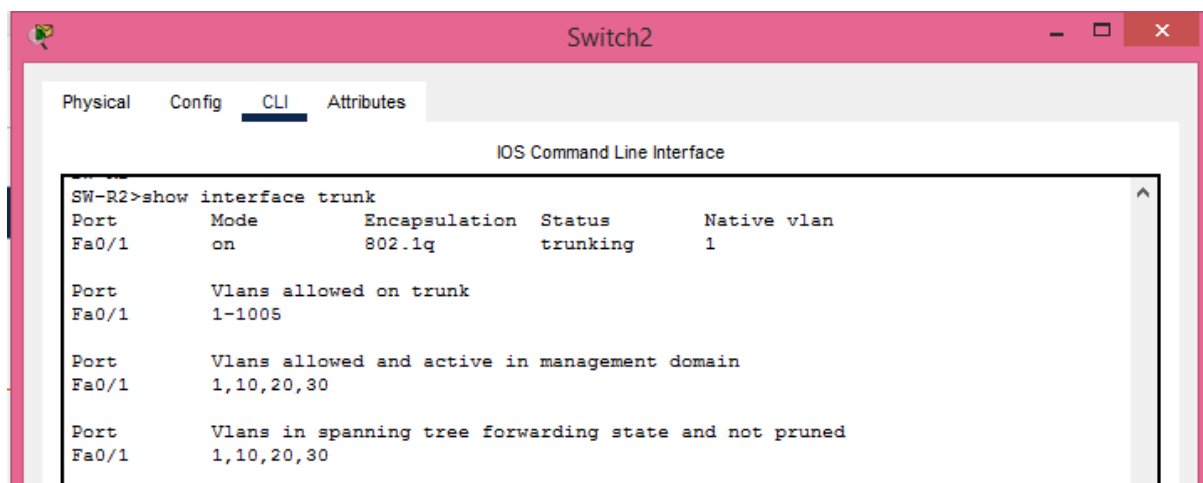
IOS Command Line Interface

```
SW-R2>show interface trunk
Port      Mode      Encapsulation  Status      Native vlan
Fa0/1     on        802.1q         trunking    1

Port      Vlans allowed on trunk
Fa0/1     1-1005

Port      Vlans allowed and active in management domain
Fa0/1     1,10,20,30

Port      Vlans in spanning tree forwarding state and not pruned
Fa0/1     1,10,20,30
```



Switch2

Physical Config CLI Attributes

IOS Command Line Interface

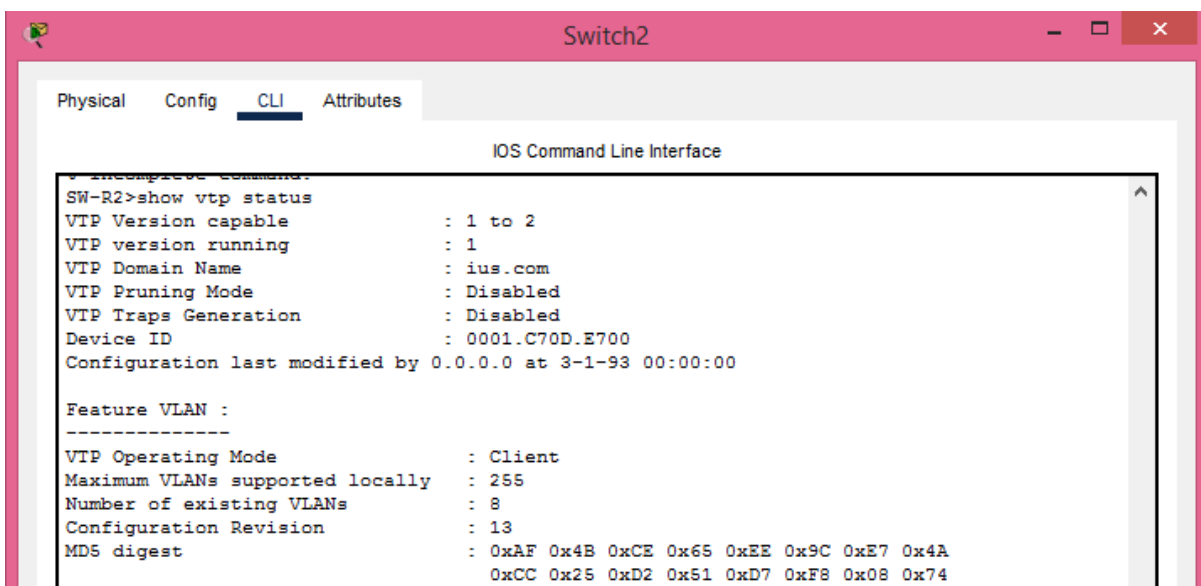
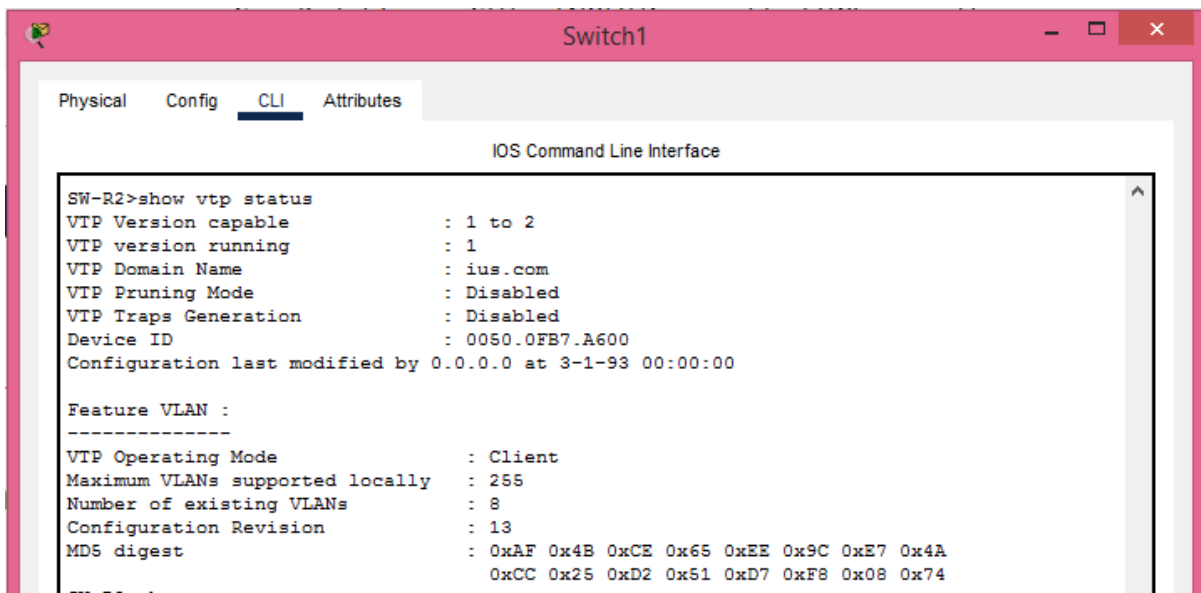
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Fa0/1     on        802.1q         trunking    1

Port      Vlans allowed on trunk
Fa0/1     1-1005

Port      Vlans allowed and active in management domain
Fa0/1     1,10,20,30

Port      Vlans in spanning tree forwarding state and not pruned
Fa0/1     1,10,20,30
```

- Vérification VTP :



- Vérification STP :

Switch1

Physical Config CLI Attributes

IOS Command Line Interface

```

4 vlans                0      0      0      8      8

SW-R2>show spanning-tree summary
Switch is in pvst mode
Root bridge for:
Extended system ID      is enabled
Portfast Default        is disabled
PortFast BPDU Guard Default is disabled
Portfast BPDU Filter Default is disabled
Loopguard Default       is disabled
EtherChannel misconfig guard is disabled
UplinkFast              is disabled
BackboneFast            is disabled
Configured Pathcost method used is short

```

Name	Blocking	Listening	Learning	Forwarding	STP Active
VLAN0001	0	0	0	1	1
VLAN0010	0	0	0	1	1
VLAN0020	0	0	0	5	5
VLAN0030	0	0	0	1	1
4 vlans	0	0	0	8	8

Switch2

Physical Config CLI Attributes

IOS Command Line Interface

```

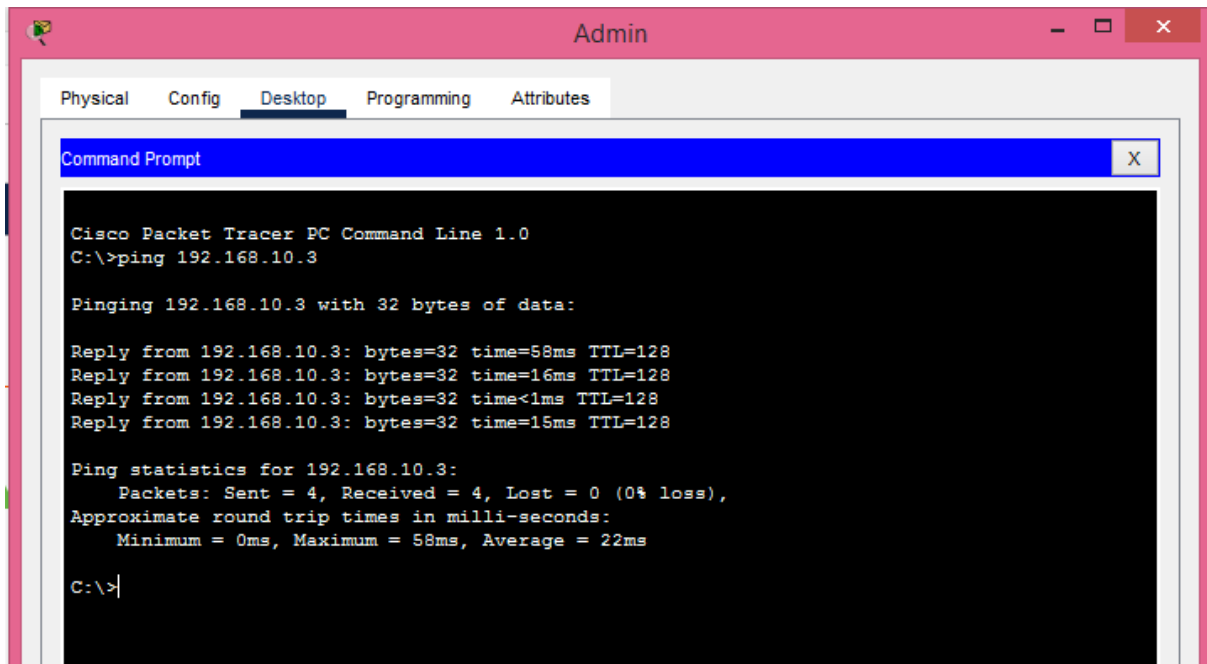
SW-R2>show spanning-tree summary
Switch is in pvst mode
Root bridge for:
Extended system ID      is enabled
Portfast Default        is disabled
PortFast BPDU Guard Default is disabled
Portfast BPDU Filter Default is disabled
Loopguard Default       is disabled
EtherChannel misconfig guard is disabled
UplinkFast              is disabled
BackboneFast            is disabled
Configured Pathcost method used is short

```

Name	Blocking	Listening	Learning	Forwarding	STP Active
VLAN0001	0	0	0	1	1
VLAN0010	0	0	0	2	2
VLAN0020	0	0	0	1	1
VLAN0030	0	0	0	3	3
4 vlans	0	0	0	7	7

Test de connectivite sur :

- VLAN 10 :



The screenshot shows a window titled "Admin" with tabs for Physical, Config, Desktop, Programming, and Attributes. The "Desktop" tab is active, displaying a "Command Prompt" window. The command prompt shows the output of a ping command to 192.168.10.3, indicating successful connectivity with 0% loss.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.10.3

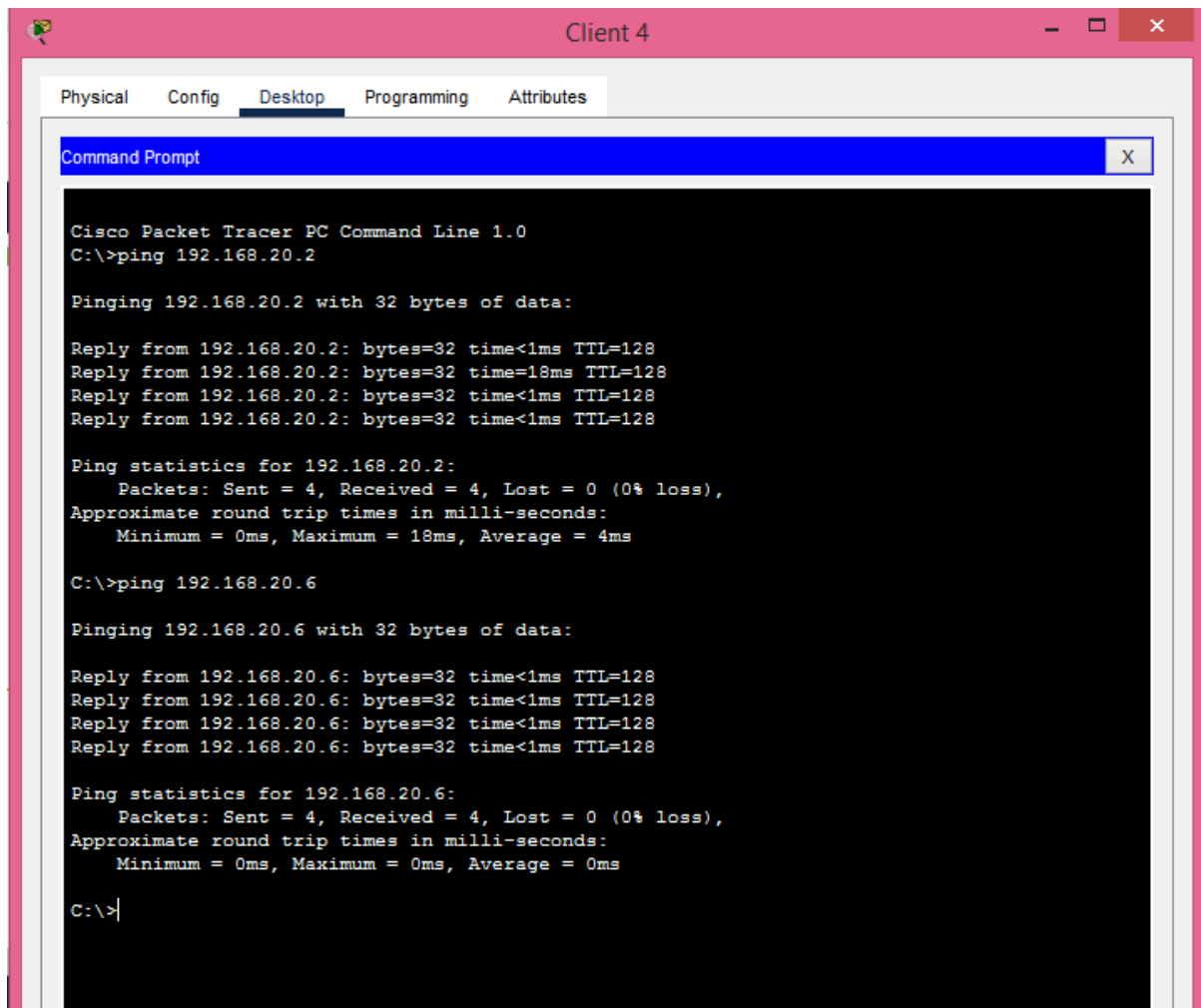
Pinging 192.168.10.3 with 32 bytes of data:

Reply from 192.168.10.3: bytes=32 time=58ms TTL=128
Reply from 192.168.10.3: bytes=32 time=16ms TTL=128
Reply from 192.168.10.3: bytes=32 time<1ms TTL=128
Reply from 192.168.10.3: bytes=32 time=15ms TTL=128

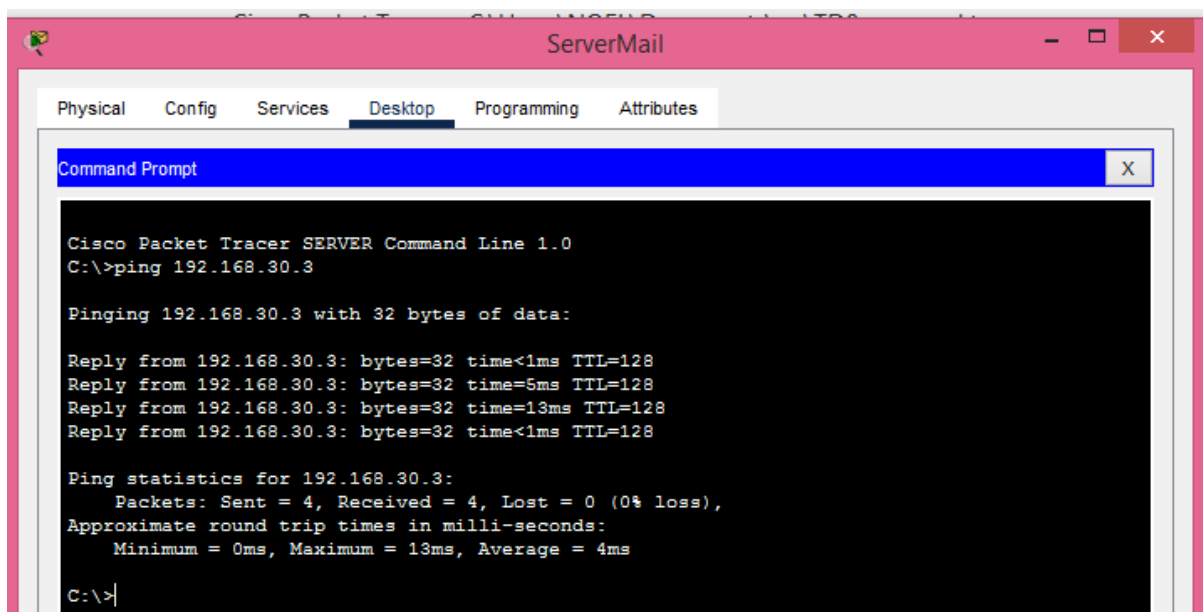
Ping statistics for 192.168.10.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 58ms, Average = 22ms

C:\>|
```

- VLAN 20 :



- Vlan 30 :



Conclusion

Ce laboratoire de réseau m' a permis d'acquérir les compétences essentielles pour segmenter et gérer un réseau à l'aide des VLAN sur Cisco Packet Tracer, à créer des VLAN, affecter des ports, configurer des trunks pour transporter plusieurs VLAN sur une seule liaison, et mettre en place le routage inter-VLAN via un routeur. Enfin, la validation des communications intra-VLAN et inter-VLAN a confirmé la bonne configuration du réseau, renforçant ainsi la compréhension des principes de segmentation et de connectivité. Une expérience pratique clé pour concevoir des réseaux efficaces et sécurisés.